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README

Machine Learning for Public Infrastructure Reliability: Predicting Water Well Status in Tanzania

Project Overview

Access to clean and reliable water remains a major challenge in Tanzania, where many water wells are either non-functional or in need of repair.

This project aims to build a **machine learning model** that predicts the condition of a water well based on its characteristics, such as location, construction details, and water quality.

The solution is designed to support **government agencies and NGOs** in identifying wells that require maintenance, enabling more efficient allocation of resources.

Objectives

- Predict the condition of a water well:
  - Functional
  - Functional but needs repair
  - Non-functional
- Identify key factors that influence well performance
- Provide actionable insights for infrastructure planning and maintenance

Dataset Description

The dataset contains information about **59,400 water wells** in Tanzania, including:

- Geographic data (latitude, longitude, region)
- Technical attributes (pump type, water source)
- Operational details (population served, payment type)
- Construction information (year built, installer, funder)

The target variable is `status_group`, representing the operational condition of each well.

Data Preprocessing

Several preprocessing steps were applied to prepare the data for modeling:

Handling Missing Values

- Numerical features were imputed using the **median**
- Categorical features were imputed using the **most frequent value**

Feature Engineering

- Created a new feature `pump_age` from construction year
- Handled invalid values (e.g., construction year = 0)

Categorical Optimization

- High-cardinality features such as `installer` and `funder` were grouped into an "**Other**" category to reduce dimensionality

Feature Selection

- Removed non-informative columns such as IDs and redundant variables
- Dropped duplicate categorical representations (e.g., grouped vs. class-based features)

Machine Learning Pipeline

A full **end-to-end pipeline** was built using Scikit-learn:

- Numerical preprocessing:
  - Missing value imputation
  - Standard scaling
- Categorical preprocessing:
  - Missing value imputation
  - One-hot encoding
- Model integration using a unified pipeline structure

This ensures reproducibility and prevents data leakage.

Models Used

The following models were trained and evaluated:

- **Random Forest Classifier**
- **Extra Trees Classifier**
- **Gradient Boosting Classifier**

Model Performance

Model	Accuracy
Random Forest	80.4%
Extra Trees	79.4%
Gradient Boosting	75.9%

The **Random Forest model** achieved the best performance and was selected as the final model.

Key Insights

- **Geographic location** (latitude and longitude) is the strongest predictor of well functionality
- **Water quantity** (e.g., dry or insufficient) strongly indicates failure
- **Older wells** are more likely to be non-functional
- Wells serving larger populations may experience higher failure rates
- Infrastructure characteristics such as extraction type and waterpoint type influence performance

Limitations

- The model performs less effectively on the "**needs repair**" category due to class imbalance
- Geographic features dominate predictions, which may limit generalizability to new regions
- Further tuning and feature engineering could improve performance

Business Impact

This model can support:

- **Preventative maintenance planning**
- **Efficient allocation of repair resources**
- **Identification of high-risk regions**
- **Improved infrastructure decision-making**

Ultimately, the solution contributes to improving **access to clean and reliable water**.

Future Improvements

- Hyperparameter tuning for improved accuracy
- Advanced handling of class imbalance
- Incorporation of geospatial analysis
- Deployment as a decision-support tool

Conclusion

This project demonstrates the application of machine learning to a real-world infrastructure problem.

By leveraging data-driven insights, it is possible to identify failing water wells and support better planning and resource allocation, ultimately improving water access in Tanzania.

Releases

No releases published
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Packages

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Contributors 1

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Languages

- **Jupyter Notebook** 100.0%